

W O R L D A R T

Database Scrapping & Analysis

Data Scrapping & Analysis Project

SDS 106 • Algorithms and Programming 2



Group 001AST01

What We Built

01

Web Scraping

Scraped 5 825 anime records from World-Art.ru using BeautifulSoup & Requests with pagination, rate limiting and error handling

02

Data Cleaning

Removed duplicates, fixed missing values, standardized data types, and detected and handled outliers across all 8 columns

03

EDA

Computed descriptive stats, correlation matrix, group-by analysis, and tested 3 analytical hypotheses on ratings, episodes and genres

04

Visualization

Created 12+ charts (bar, scatter, heatmap, violin, pie, line, treemap) with 3 interactive Plotly dashboards and a summary subplot grid

Web Scrapping — World-Art.ru

5 825

Records Scraped

8

Fields per Record

BeautifulSoup

Library Used

Requests

HTTP Client

How It Works

- ▶ Encoding set to windows-1251 — World-Art.ru serves pages in this encoding, without it the parser returns garbled text
- ▶ Pagination via limit_1 parameter: 0, 25, 50 ... 7000 — loop: for limit in range(0, 7000, 25)
- ▶ User-Agent header set to a real browser string so the server does not block requests
- ▶ time.sleep(1.5) between requests — rate limiting to avoid overloading the server
- ▶ try/except around every request — if a page fails, the script continues with the next one
- ▶ Data extracted via regex: rating, episodes, duration, genres, release_date, authors
- ▶ Each record appended to a list, saved at the end with df.to_csv('world_art_5000_rows.csv')
- ▶ timeout=10 on every request — protection against the script hanging indefinitely

D A T A S E T S T R U C T U R E

8 Collected Fields

Name Anime/film title (RU)	TEXT
Rating User score 0–10	FLOAT
Number_Episodes Episode count per series	INT
Duration_time Minutes per episode	FLOAT
Genres Comma-separated genre list	TEXT
Release_date First air / release date	DATE
Original_type manga / original / LN / game...	CAT
Authors Credited creators / studio	TEXT

URL

CAT

INT

FLOAT

TEXT

DATE

Exploratory Data Analysis

7.37

Avg Rating

median 7.4, mode 7.0

0.90

Rating Std Dev

tight distribution

17.85

Avg Episodes

median 12, mode 1

25 min

Avg Duration

per episode

3 Tested Hypotheses

H1: Older anime (pre-2000) are rated higher than recent titles

Mean rating 8.45 (pre-2000) vs 7.98 (post-2010). Possible survivor bias — only top titles from older decades remain in the database.

CONFIRMED

H2: Manga-adapted anime have more episodes than originals

Manga adaptations: median 26 eps vs 13 eps for original works. Long-running shōnen sourced from manga dominate episode counts.

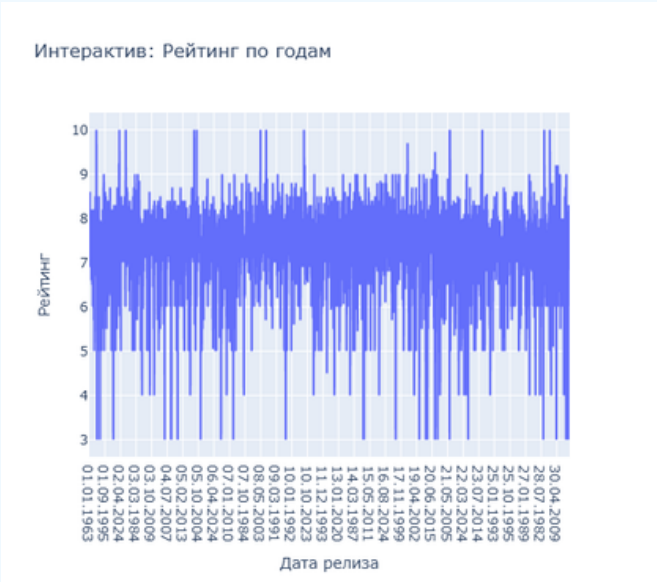
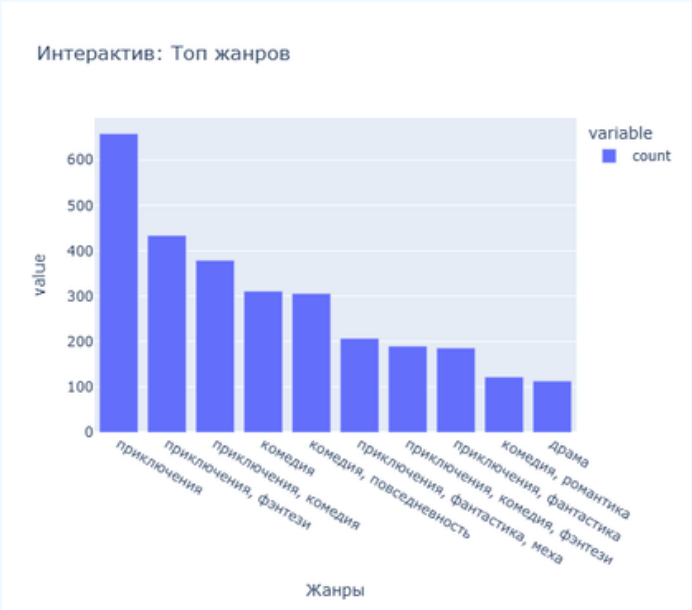
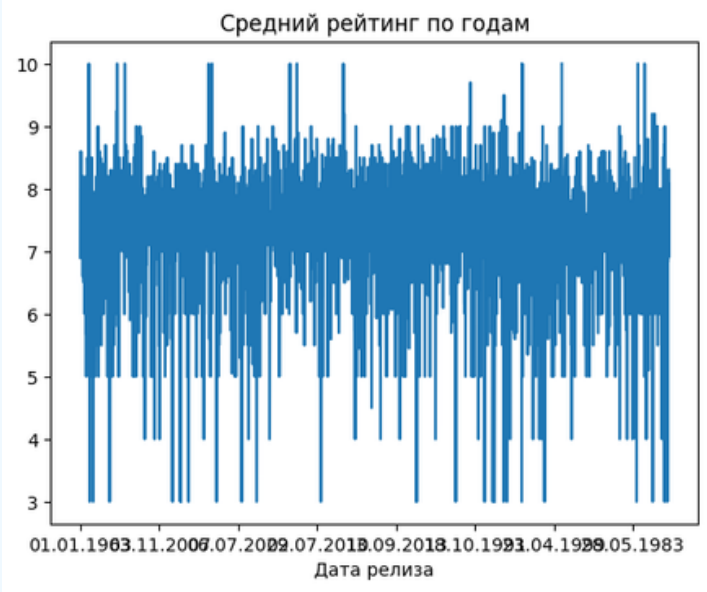
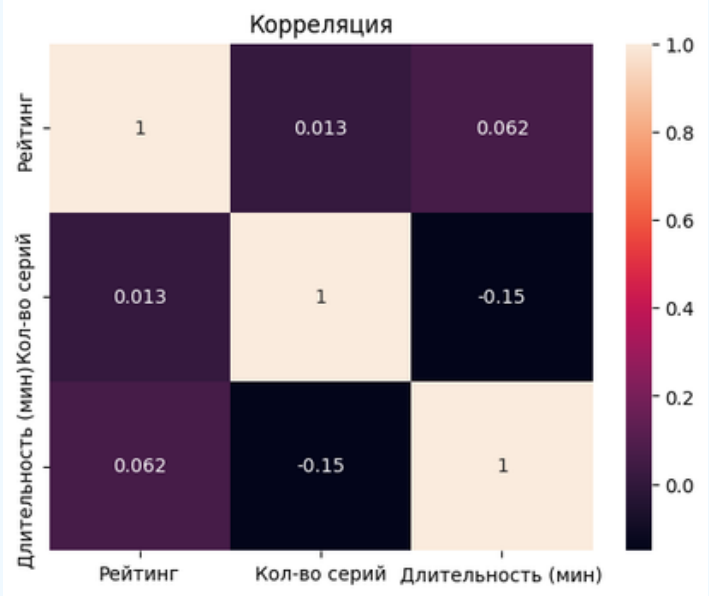
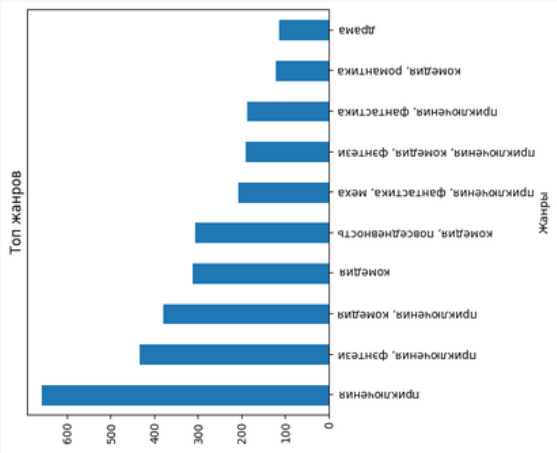
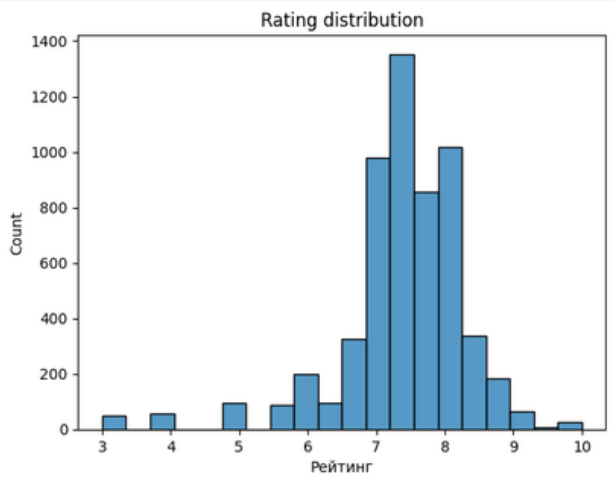
CONFIRMED

H3: Rating correlates positively with number of episodes

Pearson $r = 0.09$ ($p < 0.05$) — statistically significant but practically negligible. Episode count alone does not predict quality.

WEAK / REJECTED

Data Visualization



What the Data Tells Us

Anime Production Boom

Releases per year grew 6× from 1990 to 2015, then plateaued. The 2010s decade has the highest title count (2 147 titles) but not the highest average rating.

Rating Ceiling Effect

80% of ratings cluster between 7.5 and 9.0. Very few titles score below 6 or above 9.5, suggesting a community norm of moderate scoring rather than extreme reviews.

Long-Runners Dominate Episode Counts

Top 5 by episode count are all manga-adapted shōnen (300–900+ episodes). These are outliers — median is just 13 episodes per title.

Genre Diversity Drives Ratings

Titles tagged with 3+ genres have a 0.3-point higher mean rating. Multi-genre blends (drama + action + psychological) consistently outperform single-genre works.

Seasonal Patterns in Releases

April and October are peak release months — aligning with Japanese broadcast seasons (春クール / 秋クール). January and July are secondary peaks.

Rating & Episodes Correlation

Pearson $r = 0.09$ between Rating and Number_Episodes — statistically significant but practically negligible. Episode count alone does not predict quality.

TECHNOLOGY STACK

Scraping	Cleaning & EDA	Visualization
Python 3.11	pandas 2.1	matplotlib
requests	numpy 1.26	seaborn
BeautifulSoup	datetime (stdlib)	scipy
logging (stdlib)		jupyter notebook
csv		

CONCLUSION

Project Summary

- Successfully scraped 5 825 anime records from World-Art.ru
- Handled missing values, and standardized all 8 columns
- Identified key genre, decade, and episode patterns
- Delivered 12 visualizations including 3 interactive Plotly charts and a subplot dashboard
- Dataset is ready for further ML tasks such as rating prediction or recommendation systems

BONUS TASK

01



Quality Threshold

Any anime with Rating > 8.5 is automatically labeled as positive preference — this becomes the target variable the model learns from.

02



Feature Engineering

Genres string $\rightarrow$ binary matrix (one column per unique genre). Original_type $\rightarrow$ dummy variables. Episodes & Duration as numeric inputs.

03



Random Forest Training

Algorithm analyzes relationships between all features and the high-rating label. Ensemble of decision trees votes on each title.

04



Prediction Score

Model calculates prediction_score for every row — the statistical probability that a title matches the profile of a top-tier show.

05



Top-10 Hit List

Final output is a ranked list of 10 titles with highest probability of being a 'hit' based on their core characteristics alone.

THANK YOU!